Mid exam – Digital System Sample questions – MCQs

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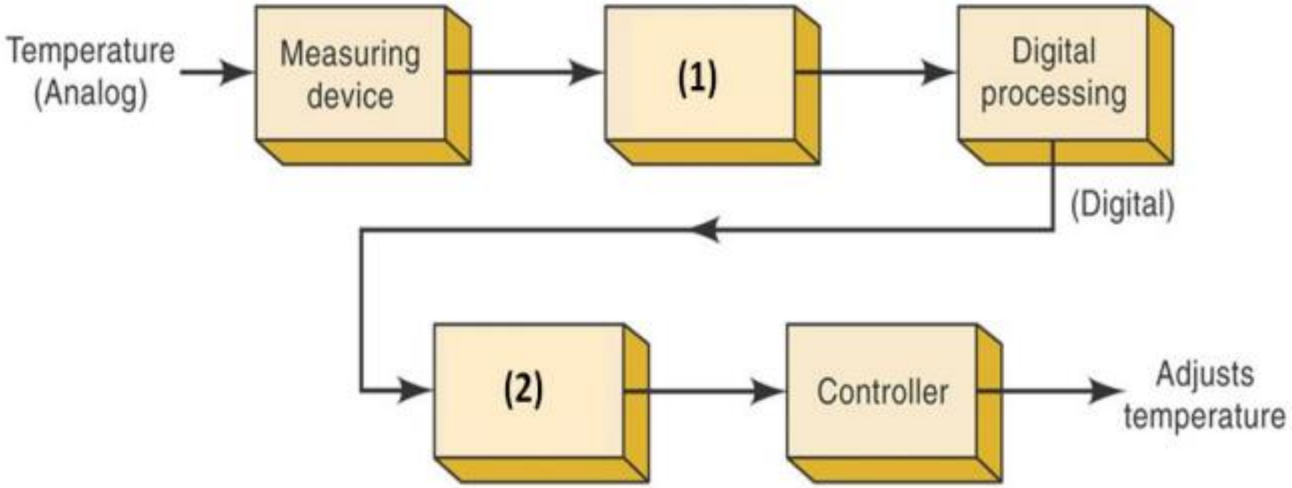
In-class exercises

Q1.1. Choose the right answer to complete the diagram for a precision temperature regulation system.



A. (2) Digital-to-analog converter – (1) Analog-to-digital converter

B. (2) Analog-to-analog converter – (1) Digital-to-digital converter

C. (2) Analog-to-digital converter – (1) Digital-to-analog converter

D. (2) Digital-to-digital converter – (1) Analog-to-analog converter

Q1.2. Which of the following is the most widely used alphanumeric code for computer input and output?

A. Gray

B. Parity

C. BCD

D. ASCII

Q1.3. Convert binary 111111110010_2 to hexadecimal.

A. $EE2_{16}$

B. $FF2_{16}$

C. $2FE_{16}$

D. $FD2_{16}$

Q1.4. How many binary digits are required to count to 100_{10}?

A. 7

B. 3

C. 2

D. 100

Q1.5. A binary number's value changes most drastically when the __________ is changed.

A. MSB

B. Frequency

C. LSB

D. Duty Cycle

Q1.6. Digital electronics is based on the _________ numbering system.

A. Decimal

B. Octal

C. Binary

D. Hexadecimal

Q1.7. The octal value corresponding to $CF.1A_{16}$?

A. 617.032

B. 613.034

C. 633.062

D. 317.064

Q1.8. Find the address range of a memory. Known that the memory size is $16Kb$, the size of each memory cell is 1 Byte, and the starting of memory address is 0:

A. $0000 - FFFF$

B. $0000 - 1FFF$

C. $0000 - 7FFF$

D. $0000 - 3FFF$

Q1.9. If a typical PC uses a 20-bit address code, the size of each memory cell is 1 Byte. How much memory can the CPU address?

A. $1MB$

B. $10MB$

C. $20MB$

D. $580MB$

Q1.10. Assign the proper odd parity bit to the code 111001

A. **1**111011

B. **1**111001

C. **0**111111

D. **0**011111

Q1.11. Assign the proper even parity bit to the code 1100001

A. 11100001

B. 1100001

C. 01100001

D. 01110101

Q1.12. The output of a logic gate is 1 when all its inputs are at logic 0. The gate is either:

A. a NAND or an EX-OR

B. an OR or an EX-NOR

C. an AND or an EX-OR

D. a NOR or an EX-NOR

Q1.13. One of the DeMorgan’s theorems shows the equivalence of:

A. OR gate and Exclusive OR gate

B. NOR gate and Bubbled AND gate

C. NOR gate and NAND gate

D. NAND gate and NOT gate

Q1.14. A universal logic gate is one, which can be used to generate any logic function. Which of the following is a universal logic gate?

A. OR

B. AND

C. XOR

D. NAND

Q1.15. The expression for Absorption law is given by __________

A. $A + AB = A$

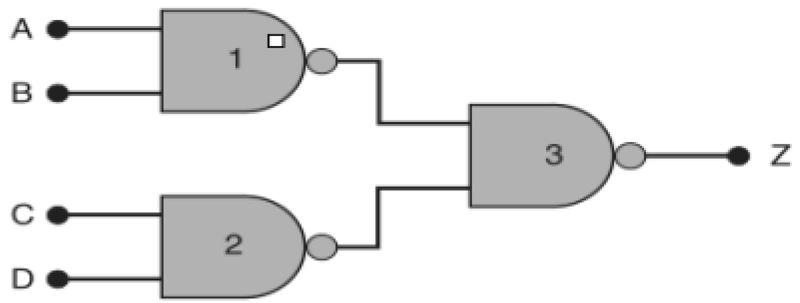
B. $A + AB = B$

C. $AB + A\bar{A} = A$

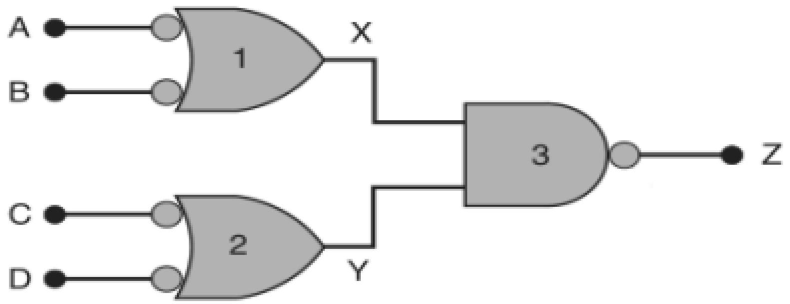
D. $A + B = B + A$

Q1.16. Which images is suitable the most with this description: “Output Z will go LOW only when A or B is LOW and C or D is LOW”.

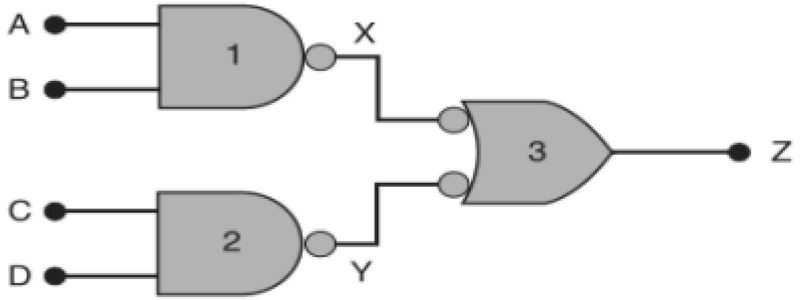
A.



C.

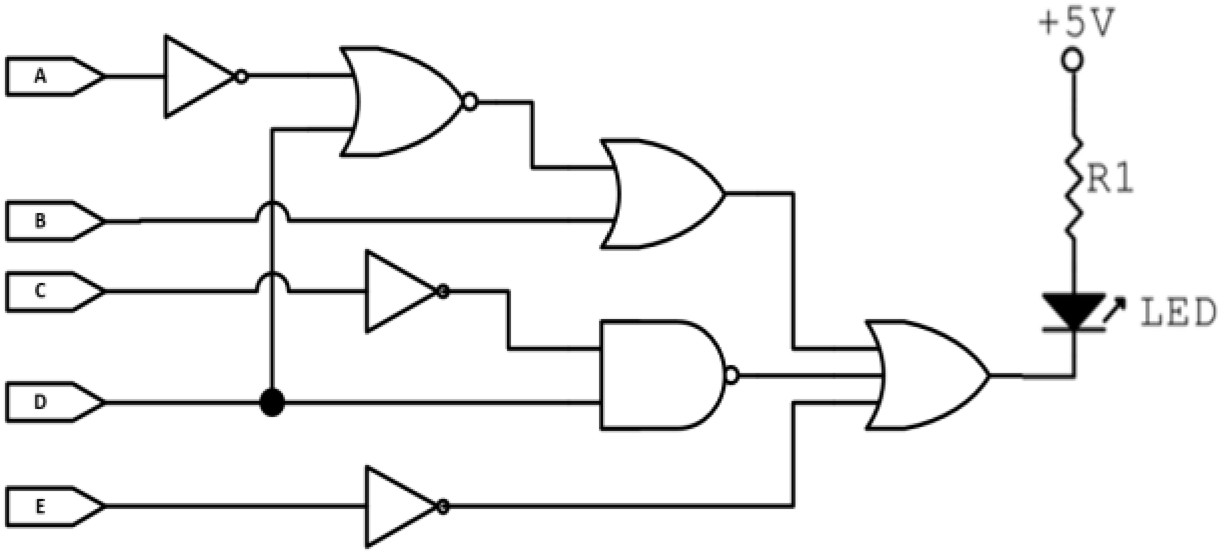


B.



D. All of them.

Q1.17. Define the combination of inputs for the circuit to make LED ON:



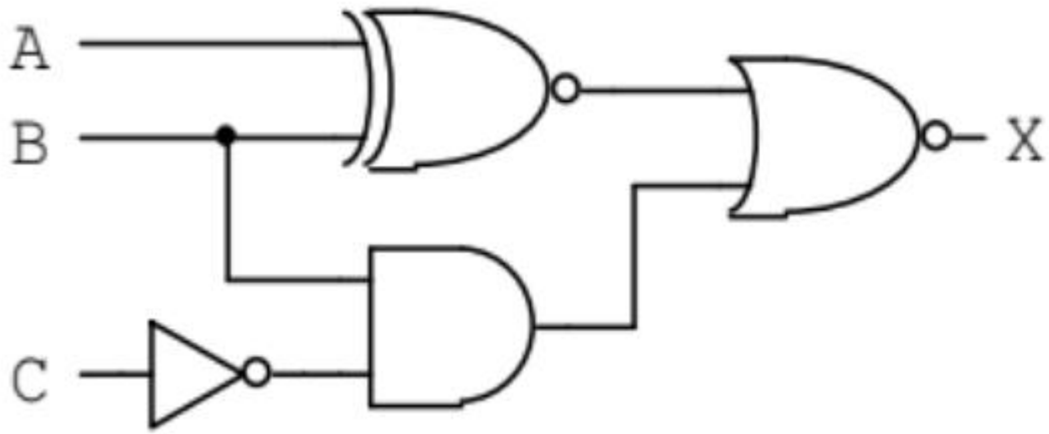
A. LED ON when $B = 0$ and $C = 0$ and $D = 1$ and $E = 1$

B. LED ON when $A = 1$ and $B = 0$ and $C = 0$ and $D = 0$ and $E = 1$

C. LED ON when $A = 0$ and $B = 0$ and $C = 0$ and $D = 1$ and $E = 0$

D. LED ON when $A = 1$ and $B = 0$ and $C = 1$ and $D = 0$ and $E = 0$

Q1.18. Find Boolean expression of the following logic circuit__________



A. $X = \overline{A\bar{B} + \bar{A}B + B\bar{C}}$

B. $X = \overline{\overline{A \oplus B} + B\bar{C}}$

C. $X = (\bar{A}\bar{B} + AB)(\bar{B} + C)$

D. All of them

Q1.19. Which of the following Boolean expressions NOT equal to F:

$$F = B\bar{C} + D\left(\overline{\bar{B} + \bar{C}} + A\bar{B}\right) + \overline{A + B + C}$$

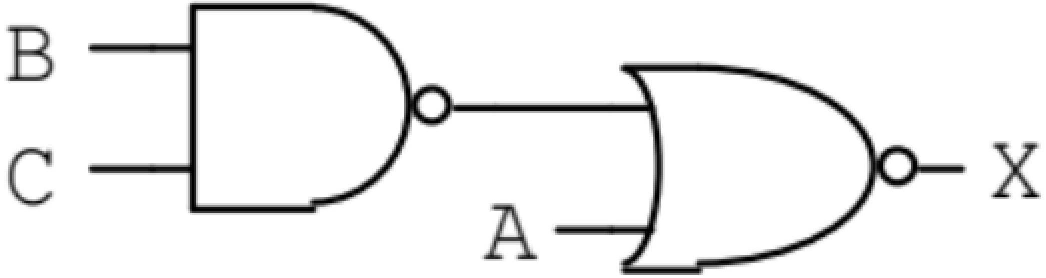
A. $B\bar{C} + BCD + A\bar{C} + A\bar{B}D$

B. $B\bar{C} + BD + \bar{A}\bar{B}\bar{C} + A\bar{B}D$

C. $B\bar{C} + BD + \bar{A}\bar{C} + AD$

D. $B\bar{C} + BCD + \bar{A}\bar{B}\bar{C} + A\bar{B}D$

Q1.20. Choose the TRUE statement:



A. $X = \bar{A}$ when $B = 1$ or $C = 1$, otherwise $X = 1$

B. $X = A$ when $B = 0$ and $C = 0$, otherwise $X = 0$

C. $X = \bar{A}$ when $B = 1$ and $C = 1$, otherwise $X = 1$

D. All is not correct

Q1.21. Given $F = \bar{A}B + \overline{B + D} \cdot \bar{B} \cdot D$, the reverse expression of F is:

A. $\bar{F} = A\bar{B}$

B. $\bar{F} = A + \bar{B}$

C. $\bar{F} = AB$

D. Tất cả đều sai.

Extra exercises

Topic 1: Number System and Codes

Q2.1. (Đề MH 2025 – Môn Tin học – Chọn 1 phương án đúng) Đẳng thức nào sau đây đúng về đơn vị lưu trữ thông tin?

A. $1\ GB = 1024\ KB$

B. $1\ MB = 1024\ B$

C. $1\ TB = 1024\ GB$

D. $1\ B = 1024\ \text{bit}$

Q2.2. (Đề MH 2025 – Môn Tin học – Câu Đ/S) Cho hai dãy bit A và B:

$$A = 1001_2$$
$$B = 11_2$$

A. Dãy bit A thể hiện số 1001 ở dạng thập phân.

B. Nếu giảm bit cuối cùng dãy A đi 1 đơn vị thì giá trị của dãy bit A tương ứng trong hệ thập phân cũng giảm đi 1 đơn vị.

C. Kết quả của phép toán **A AND B OR A** là 1001**.**

Q2.3. (Đề MH 2025 – Môn Tin học – Câu Đ/S) Trong máy tính, thông tin dạng văn bản, âm thanh hay ảnh, … đều được mã hóa dưới dạng dãy bit để lưu trữ và xử lý.

A. Bảng mã Unicode là bảng mã phổ biến hiện nay được sử dụng để mã hóa văn bản.

B. Khi lấy mẫu tín hiệu âm thanh theo thời gian, người ta rời rạc hóa đồ thị liên tục dạng hình sóng thành nhiều mẫu (đoạn) rất ngắn nối tiếp nhau.

C. Mỗi màu cơ bản trong hệ màu RGB được biểu diễn bởi một dãy bit có độ dài khác nhau.

D. Trong bảng mã ASCII mở rộng, biết ký tự A có mã thập phân là 65, vậy ký tự D có mã thập phân là 70.

Topic 2: Minimum number of 2-input NAND / NOR gates to be used to implement a logic expression F

1. NAND / NOR are **universal logic gates**!!! [Slide 16].

2. General method? → No. It belongs to the NP-hard problem

Source: https://cs.stackexchange.com/questions/96354/trick-insight-to-i-implement-given-boolean-function-with-minimum-numbers-of-give

3. We can do them **by hand** provided that the answer to the problem is not too large.

- Sure! We need to reduce the logic expression first as much as possible

→ Reduce the number of operands and operators!!!

- If it is large --> Must use a tool to support solving the problem (No exam).

Q2.4. (Important!!!) Fill in the table to show the minimum number of 2-input NAND gates and the minimum number of 2-input NOR gates required to implement some basic gates.

Basic gates	Minimum number of 2-input NAND gates	Minimum number of 2-input NOR gates
NOT		
AND		
OR		
XOR		
XNOR		
NAND		
NOR		

Q2.5. The minimum number of 2-input NAND gates required to implement the logic expression $F = \bar{A}BC$ is:

A. 3

B. 5

C. 6

D. 4

Q2.6. The minimum number of 2-input NAND gates required to implement the logic expression $F = (\bar{A} + \bar{B})(C + D)$ is:

A. 3

B. 4

C. 5

D. 6

Q2.7. The minimum number of 2-input NOR gates required to implement the logic expression $F = AB + C$ is:

A. 2

B. 3

C. 4

D. 5

Q2.8. The Boolean function is given below:

$$F = A \oplus B \oplus AB$$

Which statement(s) is/are correct?

A. It is an OR gate.

B. It is a NAND gate.

C. It requires at least 3 2-input NAND gates to implement the function.

D. It requires at least 2 2-input NOR gates to implement the function.

Topic 3: Simplified expression and Karnaugh map (K-map)

Q2.9. Which of the following functions implements the Karnaugh map shown below?

A. $\bar{A}B + CD$.

B. $D(C + A)$.

C. $AD + \bar{A}B$.

D. $(C + D)(\bar{C} + D) + (A + B)$.

$AB \backslash CD$	00	01	11	10
00	0	0	1	0
01	X	X	1	X
11	0	1	1	0
10	0	1	1	0

Q2.10. In the Karnaugh map shown below, X denotes a don't care term and all empty cells are filled with 0. What is the minimal form of the function represented by the Karnaugh map?

A. $\bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$.

B. $\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{d} + \bar{a} \cdot b \cdot \bar{d}$.

C. $\bar{b} \cdot \bar{d} + \bar{a} \cdot b \cdot \bar{d}$.

D. $\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{d} + \bar{a} \cdot \bar{d}$.

cd \ ab	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

Q2.11. The minimum sum of product expression for $f(w, x, y, z)$ is shown in the Karnaugh map below:

$yz \backslash wx$	00	01	11	10
00	0	1	1	0
01	X	0	0	1
11	X	0	0	1
10	0	1	1	X

A. $xz + \bar{y} \cdot z$.

B. $x \cdot \bar{z} + z \cdot \bar{x}$.

C. $\bar{x} \cdot y + z \cdot \bar{x}$.

D. None of the above.

Q2.12. Consider the following expression: $a \cdot \bar{d} + \bar{a} \cdot \bar{c} + b \cdot \bar{c} \cdot d$

Which of the following Karnaugh Maps correctly represents the expression?

A.

ab \ cd	00	01	11	10
00	X	X		
01	X	X		
11	X	X		X
10	X			X

B.

ab \ cd	00	01	11	10
00	X	X		
01	X			
11	X	X		X
10	X	X		X

C.

ab \ cd	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X			X

D.

ab \ cd	00	01	11	10
00	X	X		
01	X	X		X
11	X	X		X
10	X		X	X

Q2.13. Which functions does NOT implement the Karnaugh map given below?

A. $(w + x)y$.

B. $xy + yw$.

C. $(w + x)(\overline{w} + y)(\bar{x} + y)$.

D. None of the above.

$xy \backslash wz$	00	01	11	10
00	0	X	0	0
01	0	X	1	1
11	1	1	1	1
10	0	X	0	0

Topic 4: Logic circuits

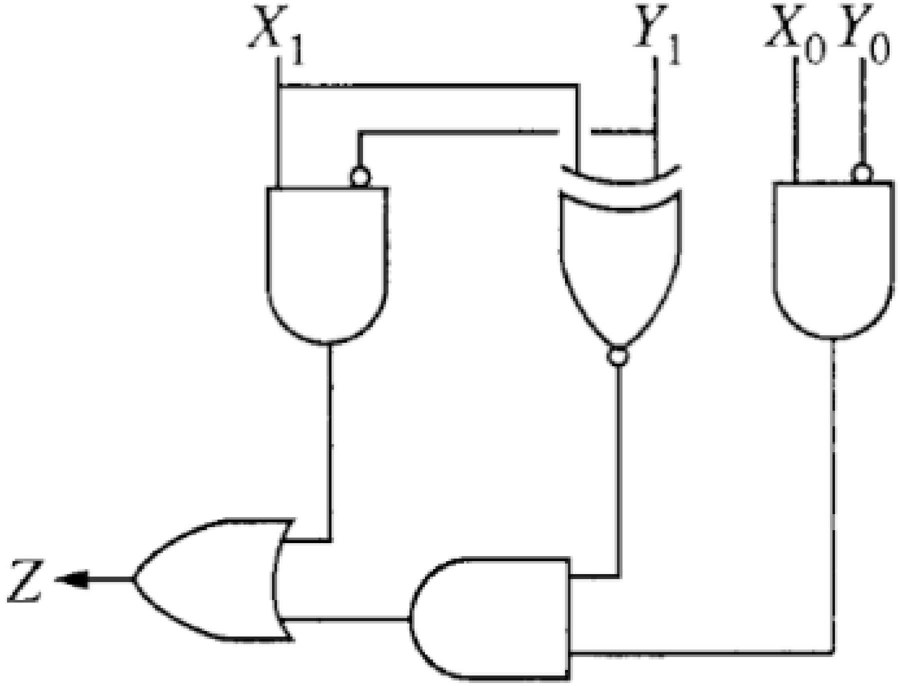
Q2.14. The logic circuit above is used to compare two 2-bit numbers, put $X = \overline{X_1 X_0}_2$, i.e X is the decimal (10-base numbers) of the binary number $X_1 X_0$ and $\mathrm{Y} = \overline{Y_1 Y_0}_2$, where X_0 and Y_0 are the least significant bits. (A small circle on any line in a logic diagram indicates logical NOT.) Which of the following always makes the output Z have the value 1?

A. $X > Y$.

B. $X < Y$.

C. $X = Y$.

D. $X \neq Y$.



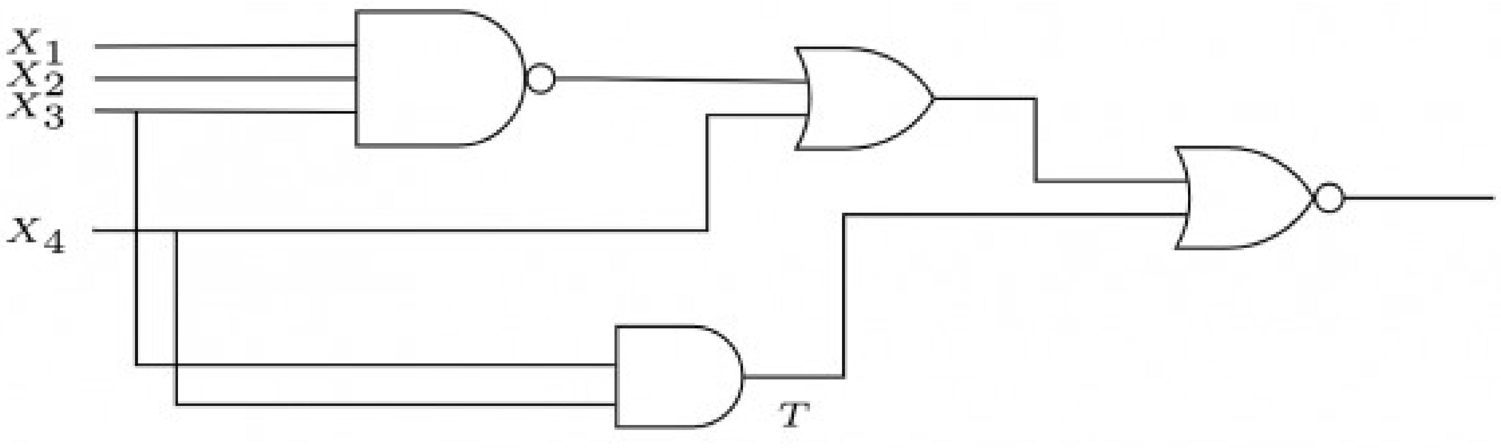
Q2.15. The line T in the following figure is permanently connected to the ground. Which of the following inputs X_1, X_2, X_3, X_4 will detect the fault?

A. $0, 0, 0, 0.$

B. $0, 1, 1, 1.$

C. $1, 1, 1, 1.$

D. None of these.



Thank you for your attention.
Good luck with your Midterm exams.